LAB TASKS

Task 01:

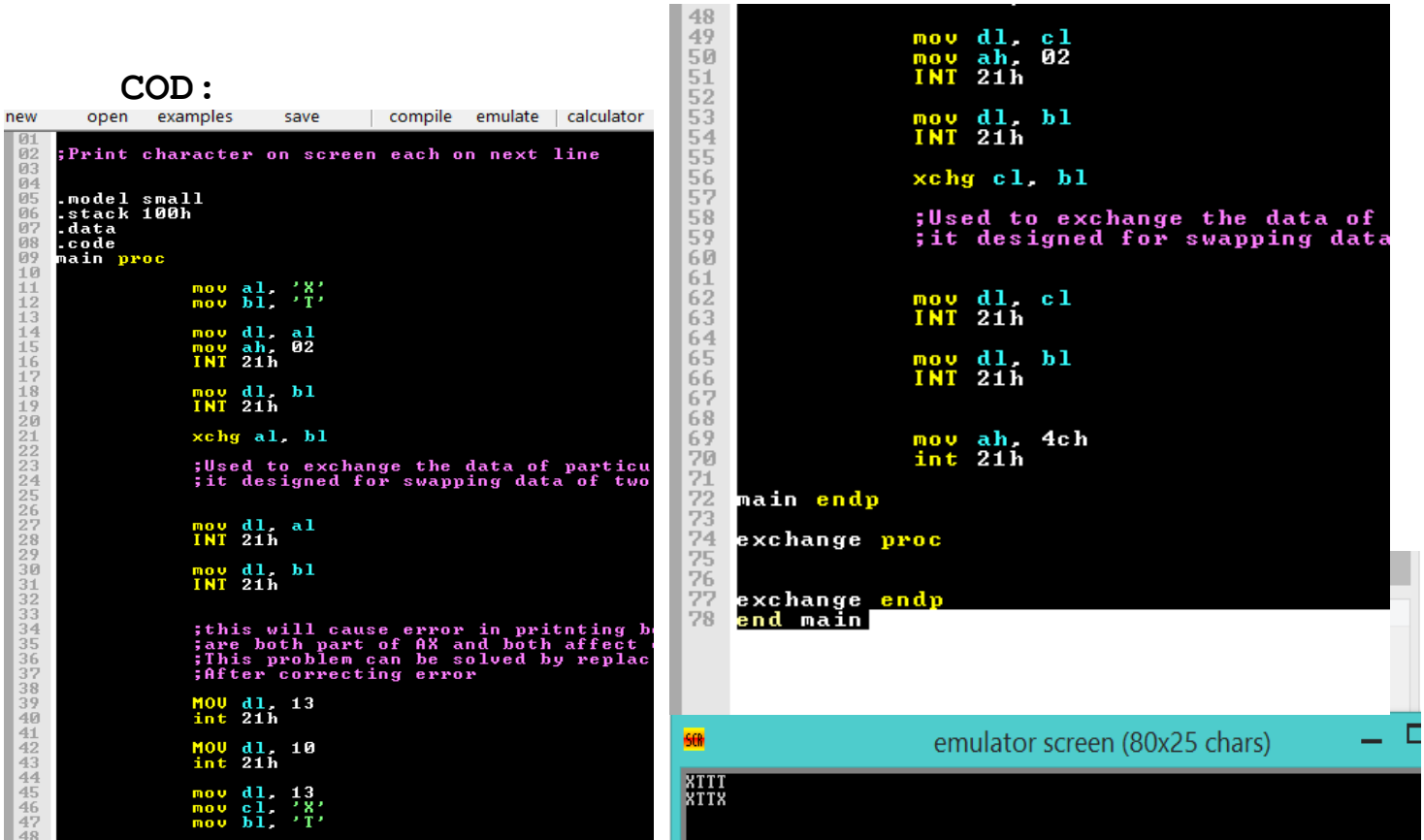
Printing IBA (each character on next line):



Task02

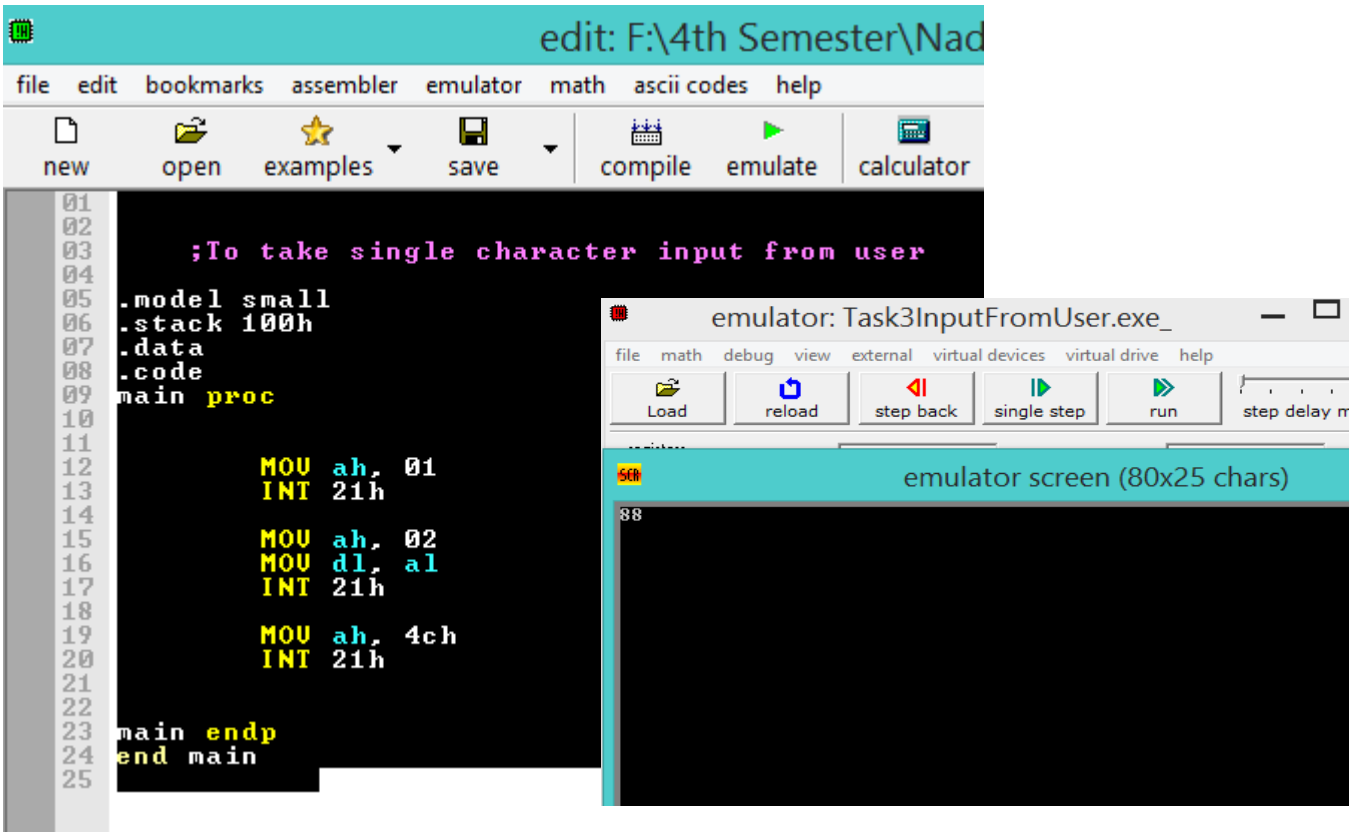
Write a program to swap two numbers or characters

COD :



Take a single character input from user and display on screen.

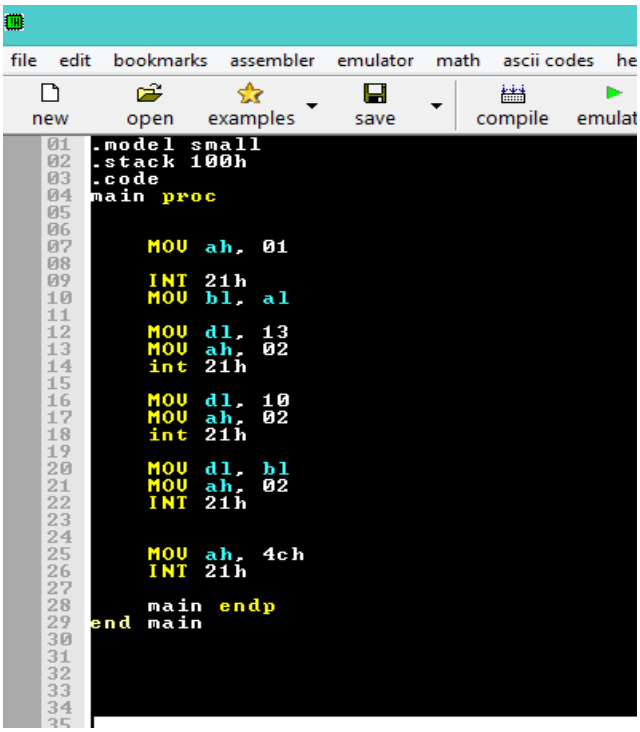
CODE:

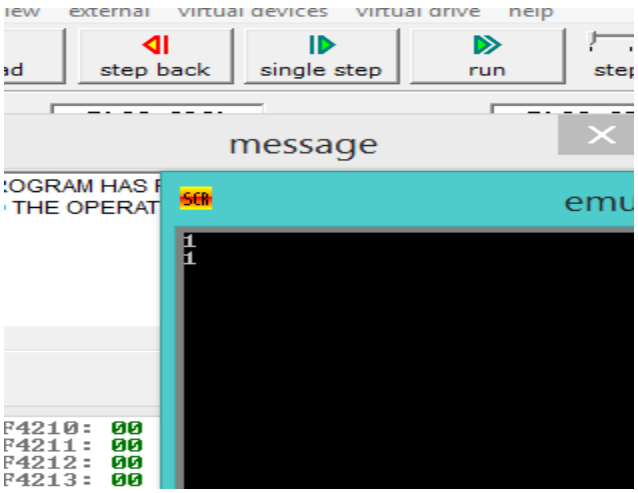


Task 04:

Take a single character input from user and display on next line.

CODE:





Task 05: ***Input character from User and print its next.***

CODE:

```
;Increment and Decrement operator
.model small
.stack 100h
.code
main proc

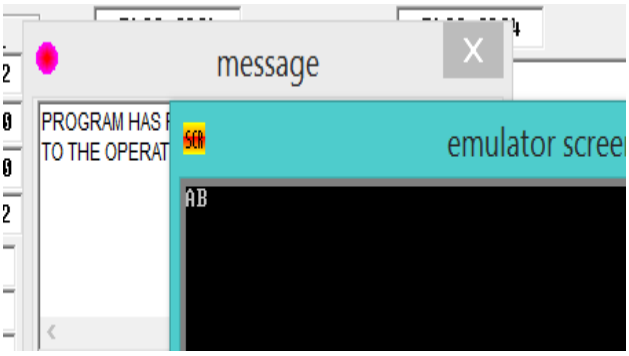

        MOV ah, 01
        INT 21h

        MOV dl, al
        INC dl
        MOV ah, 02
        INT 21h

        MOV ah, 4ch
        INT 21h

main endp
end main
```

Output:



Task 06: ***Take input from user and print its previous characters.***

```
;Increment and Decrement operator
.model small
.stack 100h
.code
main proc


        MOV ah, 01
        INT 21h

        MOV dl, al
        DEC dl
        MOV ah, 02
        INT 21h

        MOV ah, 4ch
        INT 21h

main endp
end main
```



Task 07: ***Store and Display String on screen using AL program***

```
;working with string
; We can not store string in internal memory
; Therefore, we need to store it in external

.model small
.stack 100h
.data

    str db "Hello Sir! this is my first stri

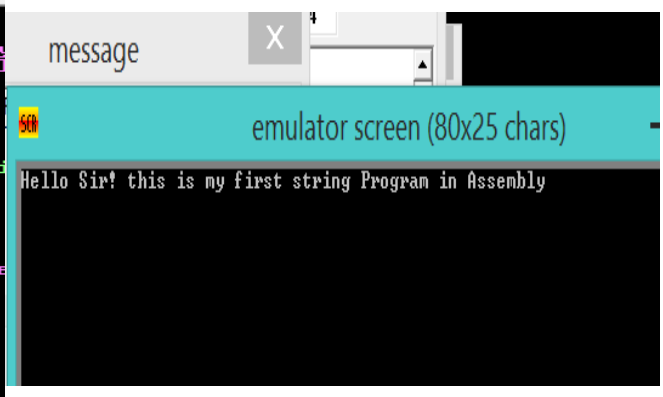
.code
main proc

        MOV AX, @data ;@ address operator
        MOV ds, Ax    ; Data segment registe

        MOV ah, 09
        lea dx, str
        INT 21h

        MOV ah, 4ch
        INT 21h


main endp
end main
```



LAB TASKS

1. Program to ask the user to enter two numbers and print the sum of those numbers.

Note: Sum must not be greater than 9.

SOURCE CODE:

Program to ask the user to enter two numbers and print the sum of those numbers.

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SOURCE CODE:

```
.model small
.stack 100h
.data
    str db "Enter two numbers for addition$"
 .code
  main proc
    mov ax,@data ;we can not direct move segment to segment
                 ;thatswhy first move into ax register,then in ds.
    mov ds,ax

    mov ah,09h
    lea dx,str ;lea(load effective address) (load address of str
               ; into dx register)
    int 21h
    ;above code will display message on screen stored in str

    mov ah,02h
    mov dl,0dh  ;caraige return
    int 21h

    mov dl,0ah   ;next line
    int 21h

    mov ah,01h  ;01h: for input from keyboard
    int 21h

    mov bl,al

    mov ah,01h
    int 21h

    add al,bl
    sub al,48

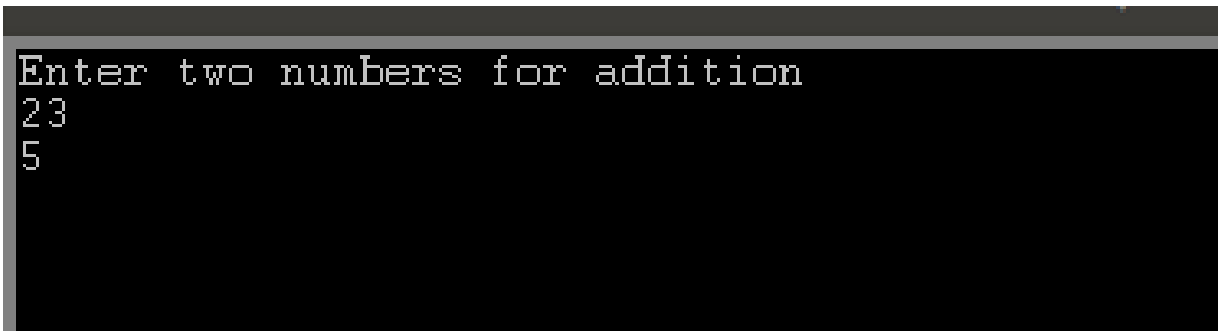
    mov cl,al  ;moving address of al into cl. because we are using
               ;ah before printing value of al.so ah affects the
               ;al register.its data is replaced when using ah.
    mov ah,02h
    mov dl,0dh  ;caraige return
    int 21h

    mov dl,0ah   ;next line
    int 21h

    mov dl,cl    ; for printing message on screen
    int 21h


    mov ah,4ch  ; 4ch S.N used for normal termination.
    int 21h


    main endp
  end main
```



TASK1:
SOURCE CODE:

```
.model small
.stack 100h
.code
main proc
   mov ax,@data
   mov ds,ax

   mov cx,26
   mov bl,65
   L1:

      mov ah,02h

      mov dl,bl
      int 21h

      inc bl

      loop L1

      mov ah,4ch
      int 21h

 main endp
end main
```

OUTPUT:



TASK:
SOURCE CODE:

```
;program to print capital latters from(A-Z) in same line and
;small letter from (z-a).every letter on next line.
.model small
.stack 100h
.data
   num db ?
.code
main proc

   mov cx,26
   mov bl,122
   mov num,65

   L1:
      mov ah,02h
      mov dl,num
      int 21h

      add num,1
    loop L1
    mov ah,02h
    mov dl,0dh
    int 21h

    mov dl,0ah
    int 21h

    mov cx,26
      L2:

     mov ah,02h

     mov dl,bl
     int 21h

     mov ah,02h
     mov dl,0dh
     int 21h

     mov dl,0ah
     int 21h

     dec bl
     loop L2

     mov ah,4ch
     int 21h

 main endp
end main
```

